

Regime Structure and Entropic Flow Ontology in Discrete Gravity Models

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Abstract

We formalise the regime structure underlying discrete entropic gravity models based on graph-defined functionals. Rather than treating gravitational dynamics as universally geometric, we show that distinct physical behaviours emerge from dominance transitions between gradient, bulk, and source sectors of a discrete entropic functional. The ultraviolet (strong-field), infrared (weak-field), and cosmological regimes arise naturally as scale-dependent responses of an entropic flow network. We clarify the ontological status of entropy, energy flow, and the cosmological constant within this framework, and demonstrate that Λ functions as a bulk capacity parameter governing large-scale screening. The analysis provides a regime-based foundation for discrete gravity models without assuming a fundamental spacetime metric.

1 Introduction

Discrete gravity models constructed from graph-based functionals have recently been shown to reproduce Newtonian behaviour in strong-field limits and MOND-like scaling in weak-field regimes [1]. Cosmological viability under Λ -locked screening has been established separately [2]. These results suggest that gravitational phenomenology may emerge from entropic flow dynamics defined on discrete networks rather than from a fundamental continuous metric structure.

The purpose of this work is not to introduce new phenomenology, but to clarify the regime structure and ontological commitments implicit in such models. We identify the mathematical conditions under which different gravitational behaviours emerge and show that these regimes correspond to dominance transitions between sectors of the discrete functional.

This paper draws on the AQUAL formulation of modified Newtonian dynamics [4], entropic gravity [5], and observational constraints from Planck [6]. For a comprehensive review of modified gravity theories, see [7].

2 Discrete Entropic Functional

Consider a cubic lattice or general graph with nodes i and adjacency structure $\langle i, j \rangle$. Define the discrete potential Φ_i and the functional

$$F[\Phi] = \sum_{\langle i, j \rangle} \kappa_{ij} \mathcal{F}\left(\frac{|\Phi_i - \Phi_j|}{a_0}\right) + \sum_i \frac{1}{2} \mu_\Lambda (\Phi_i - \Phi_V)^2 + \sum_i \lambda_i \rho_i \Phi_i, \quad (1)$$

where $\mathcal{F}(x) = \int_0^x y \mu(y) dy$ is the AQUAL kinetic integrand [4] and $\mu(x) = x/\sqrt{1+x^2}$ is the interpolating function. The three sectors are:

- **Gradient sector (UV):** nearest-neighbour coupling with nonlinear conductivity $\mu(x)$. Encodes local field dynamics.
- **Bulk sector (IR):** quadratic coupling to a volume-averaged background potential Φ_V with strength $\mu_\Lambda \propto \Lambda$. Encodes cosmological screening.

- **Source sector:** linear coupling to local matter density ρ_i . Encodes gravitational sourcing.

Stationary configurations satisfy the discrete Euler–Lagrange equation

$$\sum_{j \sim i} \kappa_{ij} \mu \left(\frac{|\Phi_i - \Phi_j|}{a_0 a} \right) (\Phi_j - \Phi_i) - \mu_\Lambda (\Phi_i - \Phi_V) = 4\pi G \rho_i, \quad (2)$$

where a is the lattice spacing. In the continuum limit, this reduces to

$$\nabla \cdot \left[\mu \left(\frac{|\nabla \Phi|}{a_0} \right) \nabla \Phi \right] - \mu_\Lambda (\Phi - \Phi_V) = 4\pi G \rho. \quad (3)$$

3 Emergent Regimes

Distinct physical regimes arise depending on which sector of the functional dominates. The transitions are not imposed; they follow from the relative magnitudes of gradient, bulk, and source terms.

3.1 Ultraviolet (Strong-Field) Regime

When $|\nabla \Phi| \gg a_0$, we have $\mu(x) \rightarrow 1$ and the bulk term is negligible. The field equation reduces to

$$\nabla^2 \Phi = 4\pi G \rho, \quad (4)$$

recovering Newtonian gravity. This has been verified numerically with log–log slopes of -2.00 [1].

3.2 Infrared (Weak-Field) Regime

When $|\nabla \Phi| \ll a_0$, we have $\mu(x) \sim x$ and the gradient sector becomes nonlinear:

$$\nabla \cdot \left(\frac{|\nabla \Phi|}{a_0} \nabla \Phi \right) = 4\pi G \rho, \quad (5)$$

leading to MOND-like scaling $g \propto r^{-1}$. The discrete operator produces $g = C \sqrt{a_0 g_N}$ with a convergent prefactor $C \approx 2.32$, implying an effective acceleration scale $a_{0,\text{eff}} = C^2 a_0 \approx 5.4 a_0$ [1].

3.3 Cosmological Regime

At sufficiently large scales, the bulk term dominates over the gradient sector. For weak fields near Φ_V , the linearised equation is

$$\nabla^2 \Phi - \mu_\Lambda \Phi = 4\pi G \rho, \quad (6)$$

which is a Yukawa-type equation with screening scale

$$L_\Lambda = \frac{1}{\sqrt{\mu_\Lambda}}. \quad (7)$$

If $\mu_\Lambda \propto \Lambda$, then

$$L_\Lambda \propto \Lambda^{-1/2} \sim 4.4 \text{ Gpc}, \quad (8)$$

linking infrared gravitational structure directly to the cosmological constant. This ensures that all structure-formation scales remain screened, preserving $\sigma_8 \approx 0.811$ [2].

4 Regime Dominance Map

The three regimes can be characterised by two dimensionless ratios:

$$\xi = \frac{|\nabla\Phi|}{a_0}, \quad \eta = \frac{\mu_\Lambda \Phi}{|\nabla^2\Phi|}. \quad (9)$$

The dominance structure is:

Regime	ξ	η	Behaviour
UV (Newtonian)	$\gg 1$	$\ll 1$	$g \propto r^{-2}$
IR (MOND-like)	$\ll 1$	$\ll 1$	$g \propto r^{-1}$
Cosmological	any	$\gg 1$	Yukawa screening

The UV→IR transition occurs at $\xi \sim 1$ (i.e., $g \sim a_0$). The IR→cosmological transition occurs at $\eta \sim 1$ (i.e., scales approaching L_Λ). No regime requires a separate theoretical framework; all three emerge from the same functional.

5 Role of Λ

In this framework, Λ is not merely a background curvature parameter or a vacuum energy density. It defines the bulk entropic capacity of the network: the strength with which the graph potential is anchored to the cosmological background Φ_V .

This has two consequences:

1. The MOND transition scale $a_0 \propto c^2\sqrt{\Lambda}$ is indirectly controlled by cosmology, not introduced as an independent local constant [3].
2. The screening scale $L_\Lambda \propto \Lambda^{-1/2}$ is automatically at the Hubble scale, ensuring cosmological viability without tuning [2].

This dual role—setting both the local acceleration threshold and the global screening length—is a structural feature of the functional, not an imposed coincidence.

6 Entropic Flow and Temporal Ordering

The discrete functional (Eq. 1) does not assume a fundamental time dimension. The field equation is an equilibrium condition: $\delta F/\delta\Phi_i = 0$.

Dynamical evolution, as implemented numerically, corresponds to iterative relaxation toward stationary configurations via damped nonlinear iteration. Temporal ordering can be interpreted as successive reconfigurations of gradient energy on the network.

We note this without overstatement: the framework is compatible with time-as-emergent interpretations, but this paper does not claim to derive temporal dynamics from the functional. That remains an open problem.

7 Discussion

The regime structure identified here demonstrates that discrete entropic gravity models naturally partition into UV, IR, and cosmological domains without requiring separate theoretical frameworks for each.

The key mechanism is dominance switching between gradient energy and bulk capacity. This is analogous to how a resistor network transitions between linear (Ohmic) and nonlinear response

depending on the driving voltage, with a global capacitor (the Λ -term) providing a return path to the reference potential.

Several features of this structure merit emphasis:

- The three regimes are exhaustive: for any field configuration, exactly one sector dominates.
- The transitions are smooth, governed by the interpolating function $\mu(x)$.
- No free functions are introduced beyond $\mu(x)$ and the Λ -proportional bulk coupling.
- The entire framework is defined on graph topology; no continuum metric is assumed.

8 Conclusion

Discrete entropic gravity models possess an intrinsic regime structure governed by gradient dominance, bulk dominance, and source strength. The cosmological constant acts as a bulk capacity parameter, linking local gravitational transitions to cosmological structure.

This regime-based ontology provides a coherent conceptual foundation for discrete gravity models without assuming a fundamental spacetime metric. The companion papers [1, 2] provide the numerical verification and cosmological constraints, respectively; this paper provides the structural framework that connects them.

References

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